

DRAFT FOR MEDICAL REVIEW — NOT PUBLISHED

This article is written but **not live**, and is not linked from anywhere on the site. It cannot publish until a reviewing clinician is named: the build refuses an article marked `clinical` with no named reviewer, which is why this page exists rather than a live URL. Please read it as the page a donor will see, mark anything that is wrong or that you would word differently, and complete the sign-off on the last page.

Which blood group can receive from which

Red cell compatibility across all eight ABO and Rh groups, why plasma compatibility runs in the opposite direction, and why whole blood follows neither rule.

Intended URL	https://raktify.choudhari.ngo/learn/blood-group-compatibility
Written by	Choudhari EduHealth India Foundation
Status	draft (clinical)
Reviewer	none named — this is what is being requested

Specific questions for the reviewer

- The red cell and plasma tables are generated from the platform's compatibility matrix, which is locked as immutable reference data. Please check the printed tables against your own reference and tell us if any single cell is wrong - a wrong cell here is a patient-safety matter, not a typo.
- *Plasma runs the other way* and *Whole blood follows neither rule* are the two places a lay reader most often misreads compatibility. Please confirm both are correct and clearly put.
- Platelet compatibility is deliberately excluded (see *Platelets are not covered here*). Confirm that omission is right for a public page rather than a gap.
- The page ends by telling donors that compatibility does not make a unit safe without crossmatch and screening. Is that caveat prominent enough?

The article, as it will publish

Compatibility is the one piece of blood donation that almost everybody half knows: O is the universal donor, AB is the universal recipient. Both of those are true for red cells and both are wrong for plasma, which is the detail that makes the rest of this page worth reading.

There are eight groups, not two systems

Your group is two facts at once. The ABO system gives A, B, AB or O depending on which antigens sit on your red cells. The Rh system adds positive or negative depending on whether the D antigen is present. Together they make the eight groups every blood bank works in: O+, O−, A+, A−, B+, B−, AB+, AB−.

Transfusion compatibility is not about matching those labels. It is about whether the recipient's immune system will attack what it is given. Someone with group A blood carries antibodies against B antigens, so giving them B red cells provokes an immediate and dangerous reaction. Group O people carry antibodies against both A and B, which is why group O recipients are the most restricted of all — the mirror image of the universal-donor fact everyone remembers.

Red cells

Red cells carry the antigens, so the rule is that the donor's antigens must be ones the recipient's antibodies will tolerate.

Recipient	Can receive red cells from
O−	O−
O+	O−, O+
A−	O−, A−
A+	O−, O+, A−, A+
B−	O−, B−
B+	O−, O+, B−, B+
AB−	O−, A−, B−, AB−
AB+	all eight groups

Read down the column rather than across and the two famous facts fall out. O− appears in every row, which is what makes it the universal red cell donor and why it is the group an emergency department reaches for before a patient's own group is known. AB+ receives from every row, which makes it the universal red cell recipient — and, less comfortably for AB+ people, means their own red cells can only go to the small number of other AB recipients.

Rh negative is the stricter half of each pair. An Rh negative recipient must have Rh negative red cells; an Rh positive recipient can take either. That asymmetry is why negative groups are disproportionately in demand relative to how many negative donors there are.

Plasma runs the other way

Plasma carries the antibodies rather than the antigens, so the compatibility direction reverses completely.

Recipient	Can receive plasma from
O	O, A, B, AB
A	A, AB
B	B, AB
AB	AB

AB plasma contains no anti-A and no anti-B antibodies, which makes **AB the universal plasma donor** — the exact opposite of the red cell picture. Group O plasma is the most restricted, because it carries antibodies against both A and B.

This is not a technicality that only matters inside a laboratory. A system that recorded compatibility as a single symmetric "compatible with" fact would be confidently wrong for every plasma transfusion it touched. Raktify holds the two directions as separate reference tables for that reason.

Whole blood follows neither rule

Whole blood — the unit exactly as collected, not separated into components — contains both the donor's red cells and the donor's plasma. Both constraints therefore apply at once, and the only group that satisfies both is the patient's own.

Whole blood is same-group only. Group O whole blood is not a universal product, because its plasma carries antibodies against A and B even though its red cells carry neither antigen. The universal-donor rule people remember belongs to separated red cells, and applying it to whole blood is a real and dangerous mistake.

Most transfusion in India is of separated components rather than whole blood, which is why the component rules are the ones usually quoted.

Platelets are not covered here

Platelet compatibility follows its own rules and this page deliberately does not state them. The platform's own platelet reference data is still marked as provisional pending the medical advisor's confirmation, and a page in this library does not get to be more confident than the signed source behind it.

If you are looking for platelets for a patient, the blood bank's transfusion medicine team will decide what is suitable.

Compatible on paper is not the last step

For red cells and whole blood, a compatible group is the beginning. The blood bank still performs a **crossmatch**: a sample of the specific patient's blood is tested against the specific unit intended for them, because individual antibodies outside the ABO and Rh systems can make a correctly-grouped unit unsuitable for one particular person.

This is why a hospital cannot simply be handed a unit of a compatible group, and why "your group is in stock" is not the same as "a unit is ready for you".

What this means for a donor

Every group is needed, and the group that is scarce depends entirely on what the hospitals in one district are treating that week. The common refrain that only O– matters is wrong: a hospital's A+ patients need A+ or O+ red cells, and there are a great many more of them.

Two things are genuinely worth knowing about your own group. If you are Rh negative, your red cells are usable by both negative and positive recipients in your ABO group, and negative donors are relatively few — so a negative donor who answers an alert is unusually useful. And if you are AB, your red cells have the narrowest reach of anyone's while your plasma has the widest.

Your group also has to be verified by a laboratory before it is used to match you to a patient. Raktify shows a group you type in yourself as unverified and never matches on it, because a group remembered from an old report is wrong often enough that trusting it would eventually send a family the wrong donor.

Sources

- National Blood Transfusion Council (NBTC), Standards for Blood Banks and Blood Transfusion Services
- Raktify platform compatibility matrix, reviewed by the foundation's medical advisor and locked as immutable reference data

Reviewer sign-off

Raktify records the reviewing clinician by name on the published page, with the review date. Nothing below is optional — an unnamed review cannot be recorded.

Full name

Qualification (e.g. MD Pathology / MD Transfusion Medicine)

Medical council registration number

Verdict

- ☐ Approved as written.
- ☐ Approved with the amendments I have marked on this document.
- ☐ Not approved — see my notes.

Review date

Next review due (we default to 12 months unless you say otherwise)

Signature

Returned sign-offs are filed in docs/medical-review/ alongside the reference-data review of 10 July 2026, and the reviewer name is published on the article itself.